

LIR Source

```
require cap(relay1)
require cap(water_sensor)
set relay1 = 1
wait 500ms
read water_sensor
halt
```

6 lines, LLM-generated

compile

$O(n)$, deterministic

one statement
→ two opcodes

Bytecode

GTWAY 5	authorize relay1
GTWAY 1	authorize sensor
LIT 1	push 1
IOW 5	write relay1
WAIT 500	delay 500 ms
IOR 1	read sensor
HALT	stop

12 bytes, varint-encoded